

Phase 4 — Sync Expansion

Implementation Plan

| **For agentic workers:** Use subagent-driven-development.

Goal: Add Sync Now buttons, device identity, history + credentials sync categories, expand Menu sync settings.

Architecture: Extend existing Settings, SyncPeriod, BackgroundService, WorkManager workers. Add generated UUID for device identity. No cloud backend needed.

Tech Stack: WorkManager, Room, Orbit MVI, SharedPreferences

Task 1: Add device identity to Settings

Files:

- Modify: `core/models/src/main/kotlin/lava/models/settings/Settings.kt`
- Modify: `core/preferences/src/main/kotlin/lava/securestorage/PreferencesStorageImpl.kt`
- Modify: `core/preferences/src/main/kotlin/lava/securestorage/PreferencesStorage.kt`
- Modify: `core/data/src/main/kotlin/lava/data/impl/repository/SettingsRepositoryImpl.kt`

Steps:

1. Add `val deviceId: String = ""` to Settings data class
 2. In `PreferencesStorageImpl`, add key `"device_id"` — on first read, if empty/null, generate and persist `UUID.randomUUID().toString()`
 3. Add `getDeviceId(): String` to PreferencesStorage interface
 4. In `SettingsRepositoryImpl`, include `deviceId` in the settings flow
 5. Compile, spotless, commit
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Task 2: Add history + credentials sync periods to Settings

Files:

- Modify: core/models/src/main/kotlin/lava/models/settings/Settings.kt
- Modify: core/models/src/main/kotlin/lava/models/settings/SyncPeriod.kt (add THIRTY_MINUTES)
- Modify: core/preferences/.../PreferencesStorageImpl.kt
- Modify: core/preferences/.../PreferencesStorage.kt
- Modify: core/data/.../SettingsRepositoryImpl.kt

Steps:

1. Add `val historySyncPeriod: SyncPeriod = SyncPeriod.OFF` and `val credentialsSyncPeriod: SyncPeriod = SyncPeriod.OFF` to Settings
 2. Add THIRTY_MINUTES to SyncPeriod enum
 3. Persist via SharedPreferences keys "history_sync_period" and "credentials_sync_period"
 4. Add repository methods `setHistorySyncPeriod()` and `setCredentialsSyncPeriod()`
 5. Compile, spotless, commit
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Task 3: Add Sync Now button to Favorites + Bookmarks

Files:

- Modify: feature/favorites/src/main/kotlin/lava/favorites/FavoritesScreen.kt
- Modify: feature/favorites/src/main/kotlin/lava/favorites/FavoritesViewModel.kt
- Modify: feature/favorites/src/main/kotlin/lava/favorites/FavoritesAction.kt
- Modify: feature/bookmarks/src/main/kotlin/lava/forum/bookmarks/BookmarksScreen.kt
- Modify: feature/bookmarks/src/main/kotlin/lava/forum/bookmarks/BookmarksViewModel.kt
- Modify: feature/bookmarks/src/main/kotlin/lava/forum/bookmarks/BookmarksAction.kt

Steps:

1. Add `SyncNowClick` action to both FavoritesAction and BookmarksAction
2. Add `isSyncing: Boolean` to both states

3. In ViewModels, SyncNowClick triggers refresh + sync use cases via viewModelScope, toggles isSyncing
 4. In Screens, add a small sync icon button in the top bar (next to back button or as a trailing icon). Shows spinner when syncing
 5. Compile, spotless, commit
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Task 4: Create SyncHistoryWorker + SetHistorySyncPeriodUseCase

Files:

- Create: core/work/impl/.../workers/SyncHistoryWorker.kt
- Create: core/domain/.../SetHistorySyncPeriodUseCase.kt
- Modify: core/work/api/.../BackgroundService.kt
- Modify: core/work/impl/.../WorkBackgroundService.kt

Steps:

1. Add syncHistory(period) and syncCredentials(period) to BackgroundService interface
 2. In WorkBackgroundService, implement both methods (schedule/cancel PeriodicWorkRequest)
 3. Create SyncHistoryWorker as @HiltWorker — currently a no-op (history is local-only, worker placeholder for future backend)
 4. Create SetHistorySyncPeriodUseCase — persists setting + schedules worker
 5. Compile, spotless, commit
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Task 5: Create SyncCredentialsWorker + SetCredentialsSyncPeriodUseCase

Files:

- Create: core/work/impl/.../workers/SyncCredentialsWorker.kt
- Create: core/domain/.../SetCredentialsSyncPeriodUseCase.kt

Steps:

1. Create SyncCredentialsWorker as @HiltWorker — iterates providers via LavaTrackerSdk, calls checkAuth() for each, reports failures via NotificationService
2. Create SetCredentialsSyncPeriodUseCase — persists setting + schedules worker

3. Compile, spotless, commit

Task 6: Expand Menu sync settings UI

Files:

- Modify: `feature/menu/src/main/kotlin/lava/menu/MenuScreen.kt`
- Modify: `feature/menu/src/main/kotlin/lava/menu/MenuViewModel.kt`
- Modify: `feature/menu/src/main/kotlin/lava/menu/MenuAction.kt`

Steps:

1. Add `SetHistorySyncPeriod` and `SetCredentialsSyncPeriod` actions
 2. Add `historySyncPeriod` and `credentialsSyncPeriod` to `MenuState`
 3. Wire `SetHistorySyncPeriodUseCase` and `SetCredentialsSyncPeriodUseCase` into `ViewModel`
 4. In `MenuScreen`, expand the sync section from 2 to 4 rows:
Favorites, Bookmarks, History, Credentials — each with sync period dropdown
 5. Compile, spotless, commit
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Task 7: CI gate + CONTINUATION.md

Run spotless, constitution check, Go tests, update docs, push to mirrors.